Dr. Babasaheb Ambedkar Technological University, Lonere.

Semester Examination – May 2019

B.Tech. Programme Chemical Engineering

Sub: Process Instrumentation

Semester: VI | Date: 21/05/2019

Max. Marks: 60 | Time: 03 Hrs.

INSTRUCTIONS TO THE STUDENTS

1. All questions carry 12 marks each.
2. Attempt any **FIVE** questions.
3. Illustrate your answers with neat sketches, diagrams etc. where ever necessary.
4. Necessary data is given in the respective questions. If such data is not given, it means that the knowledge of that data is a part of the examination.
5. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly.

Q.1 Attempt the following.

a. Explain in detail the performance characteristics of an instrument. **(8M)**

b. Write a short note on Pressure Spring thermometer. **(4M)**

Q.2 Attempt the following.

a. If 5 m^3 of certain oil weighs 45 kN. Calculate the specific weight, specific gravity and mass density of the oil. **(3M)**

b. What is manometer? Describe the working of inclined type manometer with neat diagram. **(4M)**

c. Write a short note on Bellows pressure element. **(3M)**

d. Convert 101.325kN/m^2 to head in m of water at 4°C. **(2M)**

Q.3 Attempt the following.

a. Explain the principle of operation of thermocouple. **(4M)**

b. Consider a thermocouple with an emissivity 0.75 measures the temperature of gas flowing in a duct whose wall temperature is 593 K. A thermocouple indicates 833 K. the forced convective heat transfer coefficient between the gas and the thermocouple is 120 W/ m^2K. What would be the actual temperature of gas? **(3M)**

c. Write a note on Pitot tube. **(5M)**

Q.4 Attempt the following.

a. Explain in detail variable area type flow meter with the help of diagram. **(4M)**

b. What is viscometer? Explain Falling sphere viscometer and Capillary tube viscometer in detail. **(8M)**

Q.5 Describe in detail: **(2x6M)**

a. Bubble tube method

b. Simple sight glasses method of level measurement

Q.6 Define the following terms **(6x2M)**

a. Materials for thermocouples

b. Critical temperature

c. Absolute pressure

d. Thermal expansion

e. Gay-Lussac's law

f. Invar

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